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A smarter path to 100 per cent renewable energy?

Murray Pioneer, Renmark

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SEBASTIAN CALDERON

RIVERLAND communities could play an important role in South Australia's renewable energy future, with new research highlighting the scale of infrastructure required to maintain reliable electricity supplies as renewable generation continues to expand.

Riverland-based Yes Energy chief executive officer Mark Yates said energy storage and stronger interstate connections will be critical components of a reliable renewable electricity system.

His comments follow the release of Adelaide University research, examining how much storage South Australia would need if renewable generation and storage alone were relied upon to meet electricity demand during periods of low wind and solar output.

"Adding storage and interconnectivity between the states of Australia is crucial to compliment intermittent renewables if we are going to be operating on a 100 per cent renewable grid," Mr Yates said.

"We will still require a mix of coal and gas during the transition, however as we push closer to a 100 per cent renewable grid, our reliance on coal and gas will be heavily reduced."

The university study found South Australia would require a median of 4.27GW and 162GW/h of storage across the scenarios modelled, with storage requirements increasing significantly during extended periods of poor renewable generation and limited interstate support.

Mr Yates said it was important to analyse the transition within the broader history of Australia's electricity market.

"It is important to note that renewables only made an entry into the National Electricity Market in 2014," he said.

"It has only been 12 years. We need to be conscious that transitions require people and capital and take time to execute."

The National Electricity Market commenced operations on December 13, 1998, while Victoria's Yallourn coal-fired power station first came online in 1924, and is

scheduled to close in 2028.

By comparison, the first significant renewable asset connected to the National Electricity Market was the 20MW Royalla Solar Farm, in the Australian Capital Territory, in 2014.

Mr Yates said the findings should not be interpreted as evidence that renewable energy would be unable to deliver a reliable electricity supply.

"It isn't that renewables can't work, it's that we need to combine them with smarter engineering, in the short term using a small amount of backup generation during these rare weather droughts," Mr Yates said.

The Adelaide University research reached a similar conclusion, finding storage requirements could be reduced substantially if a small amount of dispatchable generation was available during the most challenging operating periods.

Senior author Pourmousavi Kani said allowing dispatchable generation to meet between 1 and 3 per cent of annual demand reduced storage requirements by as much as 78 per cent.

"This shows the value of thinking about the whole system, not just the number of batteries," Dr Kani said.

"A small amount of firm backup, used rarely, can avoid a very large amount of overbuilding in storage."

Mr Yates said battery storage had expanded rapidly across the National Electricity Market since South Australia's reserve began operating in 2017.

"The first significant, utility-scale battery storage asset added to the NEM was the Hornsdale Power Reserve in South Australia, which officially began operating on 1st December 2017," he said.

"Since the pioneering 100MW Hornsdale battery went live in late 2017, the NEM has since added approximately 5600MW of battery storage."

The Australian Energy Market Operator's latest Integrated System Plan forecasts between 18,300MW and 21,300MW of additional grid-scale battery storage capacity.



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ity will be added across the National Electricity Market by 2030.

Mr Yates said future investment would be just as important as the technology itself.

“Every investment into Infrastructure also needs to make economic sense,” he said.

“There is a significant amount of capital globally looking for a home and Australia is considered a very stable and investable place, so as a state we should be capitalising in this investment opportunity.”

The Adelaide University research concluded that while South Australia has made significant progress towards renewable generation, careful planning around storage, transmission and reliability will be required as the state continues its transition towards a high-renewable electricity system.



Yes Energy chief executive officer Mark Yates. FILE PHOTO